LOB 1036 - Geometria Analítica

Lista de exercícios 1 - Parte 1
2º semestre de 2025

1. $P(3,-2,-5)$

2. $P(2,1,9)$

3. $\begin{cases} y = 2x - 8 \\ z = \dfrac{5}{2}x - 13 \end{cases}$

4.

a) $\begin{cases} y = \dfrac{x}{2} - \dfrac{5}{2} \\ z = -2x + 5 \end{cases}$

b) $\begin{cases} y = -x + 1 \\ z = 3 \end{cases}$

5. $\begin{cases} x = \dfrac{z}{2} - \dfrac{5}{2} \\ y = \dfrac{z}{2} - \dfrac{3}{2} \end{cases}$.

6. $m = -5$

7. a) $\begin{cases} x = 1 + t \\ y = -2 \\ z = 4 \end{cases}$

b) $\begin{cases} x = 3 \\ y = 2 + t \\ z = 1 \end{cases}$

c) $\begin{cases} x = 2 \\ y = 3 \\ z = 4 = t \end{cases}$

d) $\begin{cases} \dfrac{y+1}{-1} = \dfrac{x-4}{1}; \qquad z = 2 \end{cases}$

e) $\begin{cases} x = 2 \\ \dfrac{y+3}{2} = \dfrac{z-4}{-1} \end{cases}$

8.

a) $\theta = 60º$

b) $\theta = 30º$

c) $30º$

d) $\theta = \cos^{-1}\left(\dfrac{2}{3}\right)$

9. $m = -4$

10. $m = 10, n = 5.$

11. $D(0,1,0)$

12. $m = -\frac{3}{2}$ e $m = 1.$

13.

a) $m = 4$

b) $m = -7$

c) $m = \frac{3}{2}$

14.

a) $I(1,2,3)$

b) $I(4,3,9)$

c) $I(2,1,-2)$

d) $I(1,-5,5)$

15. $\begin{cases} x = 3 - t \\ y = 2 \\ z = 1 - t \end{cases}$

16. $\begin{cases} x = 2 + t \\ y = -1 - 5t \\ z = 3t \end{cases}$

17. $a = 14, b = -10$

18.

a) $P(15,-3,33)$

b) $Q\left(5, -\frac{1}{2}, -\frac{9}{2}\right)$